



ECON-421 Project Presentation

TYSON'S REGRESSIONS

Project Overview

- ↪ Public Finance project
- ↪ Fiscal and Regulatory policies differ between U.S.
- ↪ Laboratory of Democracy is useful for data collection
- ↪ GSP (Gross State Product) and GSP attributable to the private sector must account for differences in populations between states

Data Used

- ↪ Size of state economy in real terms (real GSP) and economic output attributable to the private sector comes from the BEA (Bureau of Economic Analysis)
- ↪ State population information is from the census for years other than when the decadal census is taken, estimates are used
- ↪ Realgsp and Realgsp from private sector employment / population is used for the percapita variables, the dependent variable in the regression
- ↪ CATO institute publishes their freedom in the 50 states report, which has 230 variables relating to fiscal, regulatory and personal freedoms
- ↪ 22 variables from the fiscal and regulatory were pulled for my project
- ↪ 2 variables were excluded from the regression due to multicollinearity
- ↪ Project has 20 independent variables

Variable Explanations

- ▶ Stategovconsumptioninvestmentdiv: total state government spending for their own operations divided by the total income for a state's economy
- ▶ Localtaxpercentincome: local taxes as a percent of the total income for a state's adjusted personal income
- ▶ Statelocalemploymentdivprivateemp: total size of state and local workforce divided by the number of private sector employees. We also include government employment, which can crowd out employment in the private sector
- ▶ Sumeducexperrequirements: how many individual educational and experience requirements are there to work across all regulated occupations.
- ▶ Regcompensationassessmentrequire: Compensation required for or economic assessment required before regulatory taking (stricter than federal standard)? (=2 if both, =1 if one of the two, =0 if neither, multiplied by 0.5 if applies to state government only, not localities).
- ▶ Whartonlanduseregulationindex: This aggregate measure is comprised of eleven subindexes that summarize information on the different aspects of the regulatory environment. A higher index number means more restrictive land use regulations

First Regression Output

Fixed-effects (within) regression
Group variable: **state_n**

R-squared:
Within = **0.4531**
Between = **0.0772**
Overall = **0.0394**

corr(u_i, Xb) = **-0.8256**

Number of obs = **1,050**
Number of groups = **50**

Obs per group:
min = **21**
avg = **21.0**
max = **21**

F(20, 49) = **18.78**
Prob > F = **0.0000**

(Std. err. adjusted for **50** clusters in **state_n**)

percapitagsp	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
stategovconsumptioninvestmentdiv	-218.5047	187.8399	-1.16	0.250	-595.9832	158.9738
localtaxpercentincome	259.0908	100.1219	2.59	0.013	57.88822	460.2933
logcompetingjurisdiction	8.372929	6.589785	1.27	0.210	-4.86974	21.6156
statetaxpercentincome	-478.0211	331.3731	-1.44	0.156	-1143.94	187.8981
debtpercentincome	-2.491767	5.493517	-0.45	0.652	-13.5314	8.54787
statelocalemploymentdivrivatemp	-18.96527	39.17182	-0.48	0.630	-97.684	59.75346
sumfiscal	-52.1664	40.43401	-1.29	0.203	-133.4216	29.08879
sumlocaltax	93.64074	42.28999	2.21	0.031	8.655824	178.6257
regcompensationassessmentrequire	.068014	.5039356	0.13	0.893	-.9446825	1.08071
eminentdomainreform	-.3545926	.1878468	-1.89	0.065	-.7320849	.0228996
whartonlanduseregulationindex	.4169657	.5189553	0.80	0.426	-.6259141	1.459846
minimumtomedianwageratio	.2436417	2.321282	0.10	0.917	-4.42115	4.908433
discriminationlaw	-1.524357	.5905092	-2.58	0.013	-2.711029	-.3376839
sumhealthinsurancefreedom	3.379346	1.313641	2.57	0.013	.7394852	6.019208
sumcabletelecomm	-.3827297	.54287	-0.71	0.484	-1.473668	.7082085
sumeducexpperrequirements	.0000512	.000024	2.13	0.038	2.93e-06	.0000994
sumoccupationalfreedom	63.35795	53.1257	1.19	0.239	-43.40214	170.118
otherregulationfreedom	12.60504	25.53674	0.49	0.624	-38.71297	63.92305
nonfederalregulatorysum	-2.231386	5.103147	-0.44	0.664	-12.48654	8.023772
cronyism	-25.85218	40.24771	-0.64	0.524	-106.733	55.02862
_cons	59.83622	44.14999	1.36	0.182	-28.88651	148.5589

Second Regression Output

Fixed-effects (within) regression
Group variable: **state_n**

Number of obs = **1,050**
Number of groups = **50**

R-squared:

Within = **0.5144**
Between = **0.0798**
Overall = **0.0325**

Obs per group:

min = **21**
avg = **21.0**
max = **21**

corr(u_i, Xb) = **-0.8166**

F(20, 980) = **51.90**
Prob > F = **0.0000**

percapitaprivategsp	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
stategovconsumptioninvestmentdiv	-250.8948	128.6196	-1.95	0.051	-503.2963	1.506644
localtaxpercentincome	332.5163	95.56558	3.48	0.001	144.9796	520.053
logcompetingjursdiction	10.35011	2.816213	3.68	0.000	4.823612	15.87661
statetaxpercentincome	-589.6178	225.9345	-2.61	0.009	-1032.989	-146.2467
debtpercentincome	-3.088225	2.831366	-1.09	0.276	-8.644462	2.468012
statelocalemploymentdivprivateemp	-19.56476	26.92438	-0.73	0.468	-72.40083	33.2713
sumfiscal	-64.25129	27.32549	-2.35	0.019	-117.8745	-10.62809
sumlocaltax	118.6411	28.80352	4.12	0.000	62.11747	175.1648
regcompensationassessmentrequire	-.0380138	.2591688	-0.15	0.883	-.5466034	.4705758
eminentdomainreform	-.4767842	.064754	-7.36	0.000	-.6038567	-.3497117
whartonlanduserseregulationindex	.4766574	.2333007	2.04	0.041	.018831	.9344837
minimumtomedianwageratio	.3732411	1.400422	0.27	0.790	-2.37493	3.121412
discriminationlaw	-2.349897	.4446189	-5.29	0.000	-3.222411	-1.477382
sumhealthinsurancefreedom	5.217459	1.04531	4.99	0.000	3.166155	7.268762
sumcabletelecomm	-.4876781	.369358	-1.32	0.187	-1.212502	.2371456
sumeducexpperrequirements	.0000562	.0000429	1.31	0.191	-.0000281	.0001404
sumoccupationalfreedom	79.5912	24.47636	3.25	0.001	31.55909	127.6233
otherregulationfreedom	11.45932	14.30682	0.80	0.423	-16.61621	39.53484
nonfederalregulatorysum	-3.423091	2.924894	-1.17	0.242	-9.162867	2.316685
cronyism	-31.59865	21.38621	-1.48	0.140	-73.56669	10.36939
_cons	69.92584	31.2196	2.24	0.025	8.660889	131.1908
sigma_u	8.0841423					
sigma_e	1.4068805					
rho	.97060398	(fraction of variance due to u_i)				

Key Points

- ↪ Every percentage point increase in state government consumption and investment as a percent of GSP reduces percapitagsp and percapitaprivategsp by more than \$200
- ↪ Every percentage point increase in state tax burden reduces percapita privategsp and percapitaprivategsp by more than \$400 and \$500 respectively
- ↪ Size of public sector, regulation assessments and the minimum-to-median wage ratio are not statistically significant

Points of Concern

- ▶ Omitted Variable Bias
- ▶ 210 CATO variables not in this regression
- ▶ Factors outside of state policy